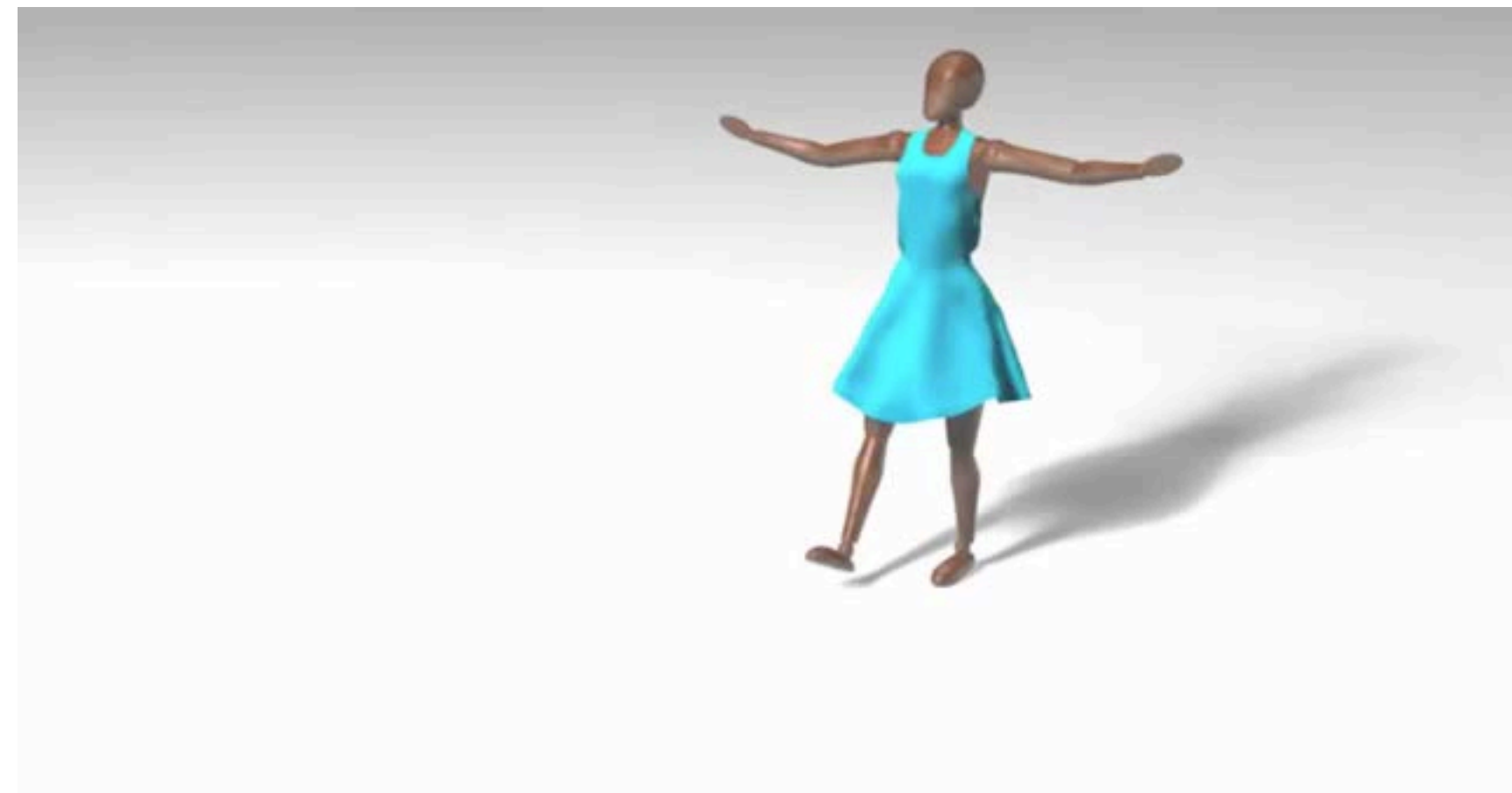


Math (P)Review Part I:

Linear Algebra

Linear Algebra in Computer Graphics

- Today's topic: **linear algebra**.
- Why is linear algebra important for computer graphics?
 - Effective bridge between geometry, physics, etc., and *computation*.
 - In many areas of graphics, once you can express the solution to a problem in terms of linear algebra, you're essentially done: now ask the computer to solve $Ax=b$.
 - Fast numerical linear algebra has really made modern computer graphics possible (image processing, physically-based animation, geometry processing...)



Vector Space—Formal Definition

- Linear algebra is the study of **vector spaces** and **linear maps** between them—here's the formal definition*:

For all vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ and scalars a, b :

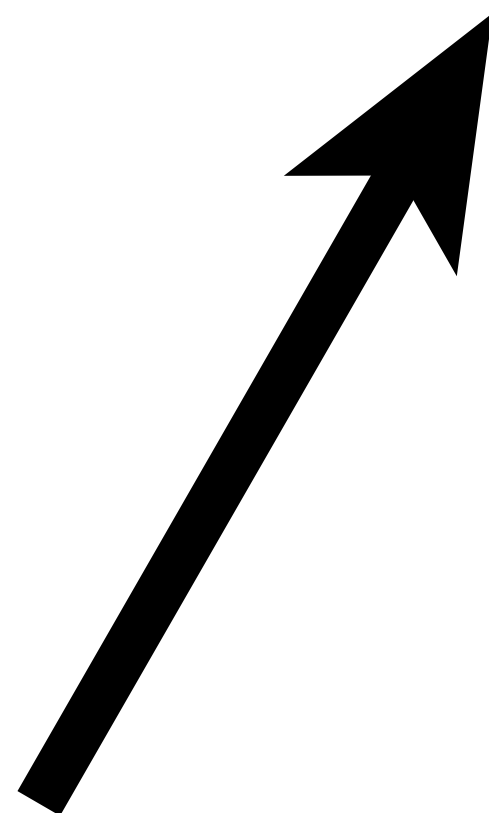
- $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
- $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$
- There exists a *zero vector* “ $\mathbf{0}$ ” such that $\mathbf{v} + \mathbf{0} = \mathbf{0} + \mathbf{v} = \mathbf{v}$
- For every $\mathbf{v}$ there is a vector “ $-\mathbf{v}$ ” such that $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$
- $1\mathbf{v} = \mathbf{v}$
- $a(b\mathbf{v}) = (ab)\mathbf{v}$
- $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$
- $(a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$

- *Where do these rules come from?*
- In mathematics (and in life) you should never simply accept a set of rules handed to you by an authority...
- Let's try to understand where these “rules” come from.

***this will NOT be on the test!**

Vectors - Intuition

- First things first: what is a **vector**?
- Intuitively, a vector is a little arrow:

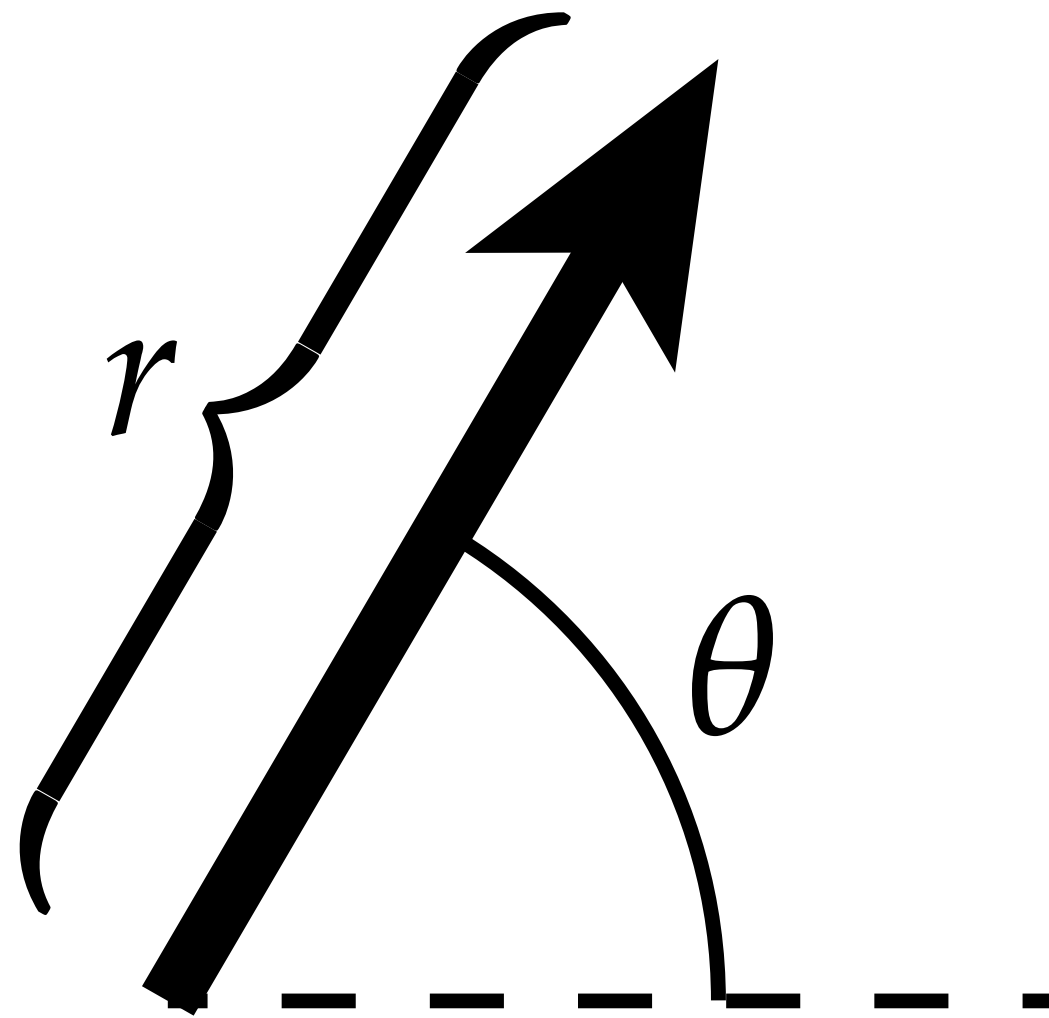


A vector.

- In computer graphics, we work with many types of data that may not look like little arrows (polynomials, images, radiance...). But they still *behave* like vectors. So, this little arrow is still often a useful mental model.

Vectors - What Can We Measure?

- What information does a vector encode?
- Fundamentally, just **direction** and **magnitude***:



- For instance, a vector in 2D can be encoded by a length and an angle relative to some fixed direction (“polar coordinates”).
- (Side note: are these values the same in any coordinate system?)
- How else might we encode a vector?

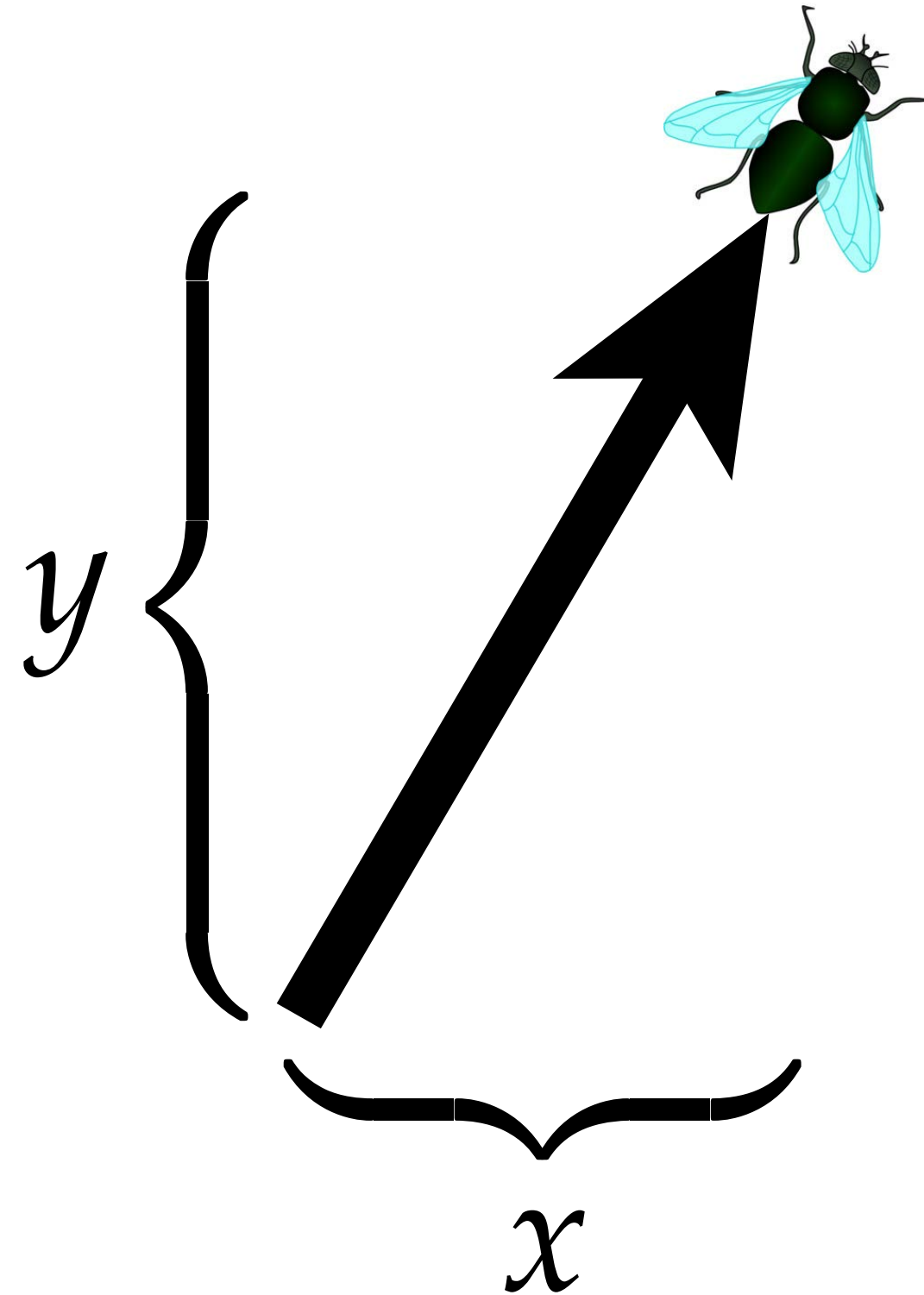
Traditionally, a vector does not include a “basepoint”; a vector with a basepoint is sometimes called a **tangent vector.*

Vector in Cartesian Coordinates

- Can also measure components of a vector with respect to some chosen *coordinate system*:



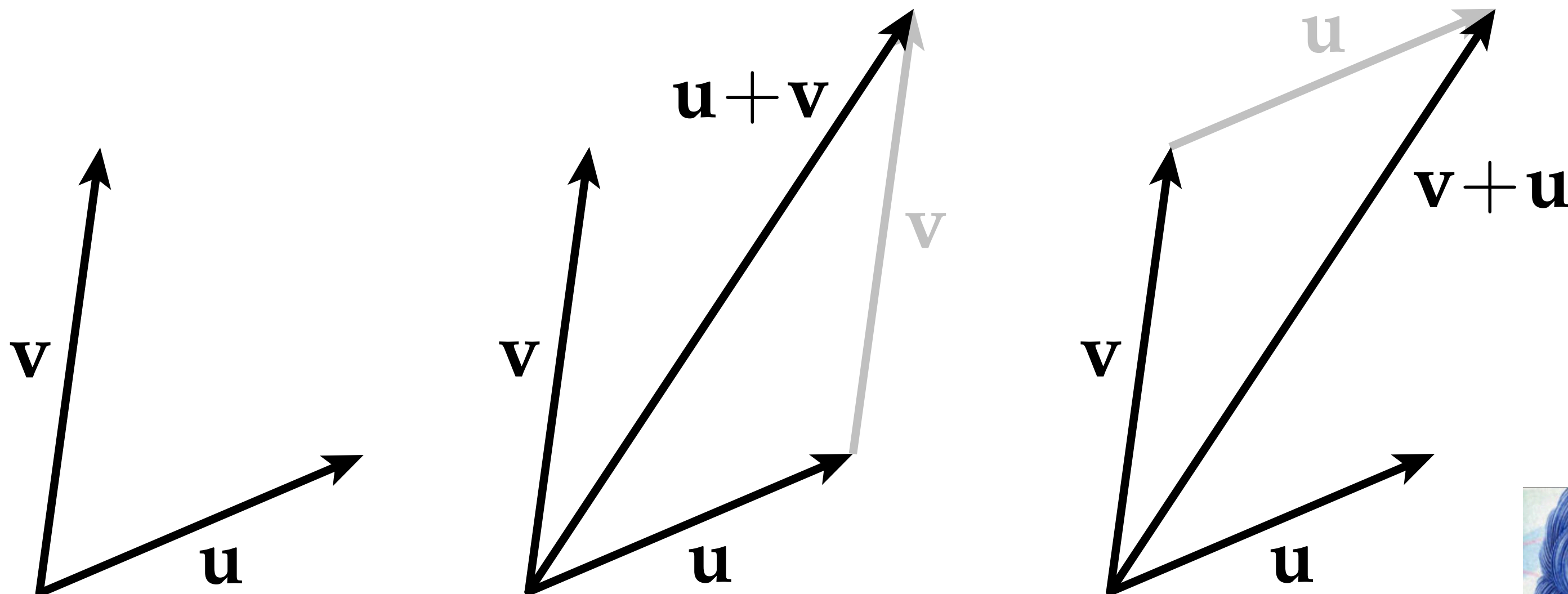
René Descartes, Est. 1596



- **WARNING:** Can't directly compare coordinates in different systems! (Also shouldn't compare (r, θ) to (x, y) .)

What Can We Do with a Vector?

- Two basic operations. First, we can add them “end to end”:



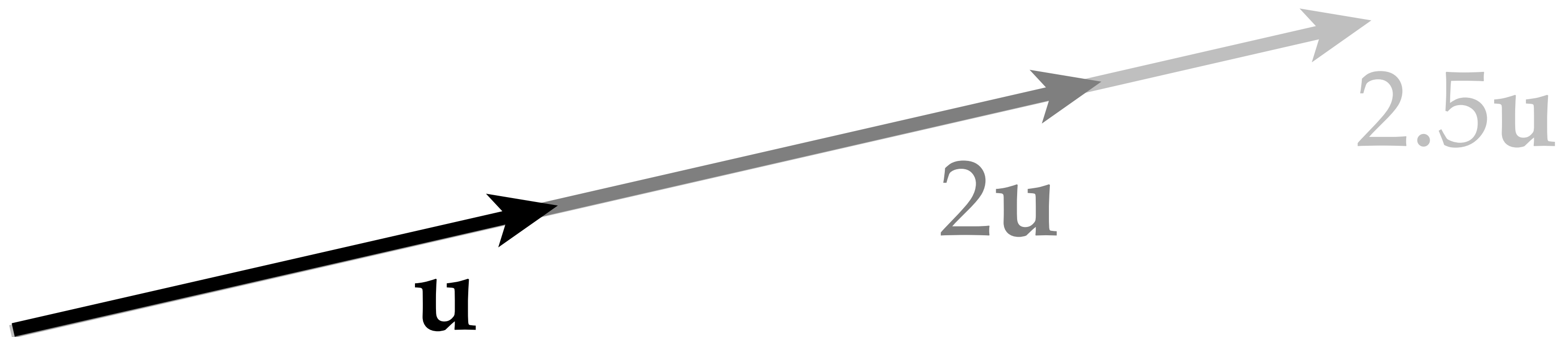
- What if we walk along v first, then u ?
- Actually, it doesn't seem to matter: $u + v = v + u$
- Language: vector addition is “commutative” or “abelian”



Niels Henrik Abel

What Else Can We Do with a Vector?

- Other basic operation? Scaling:

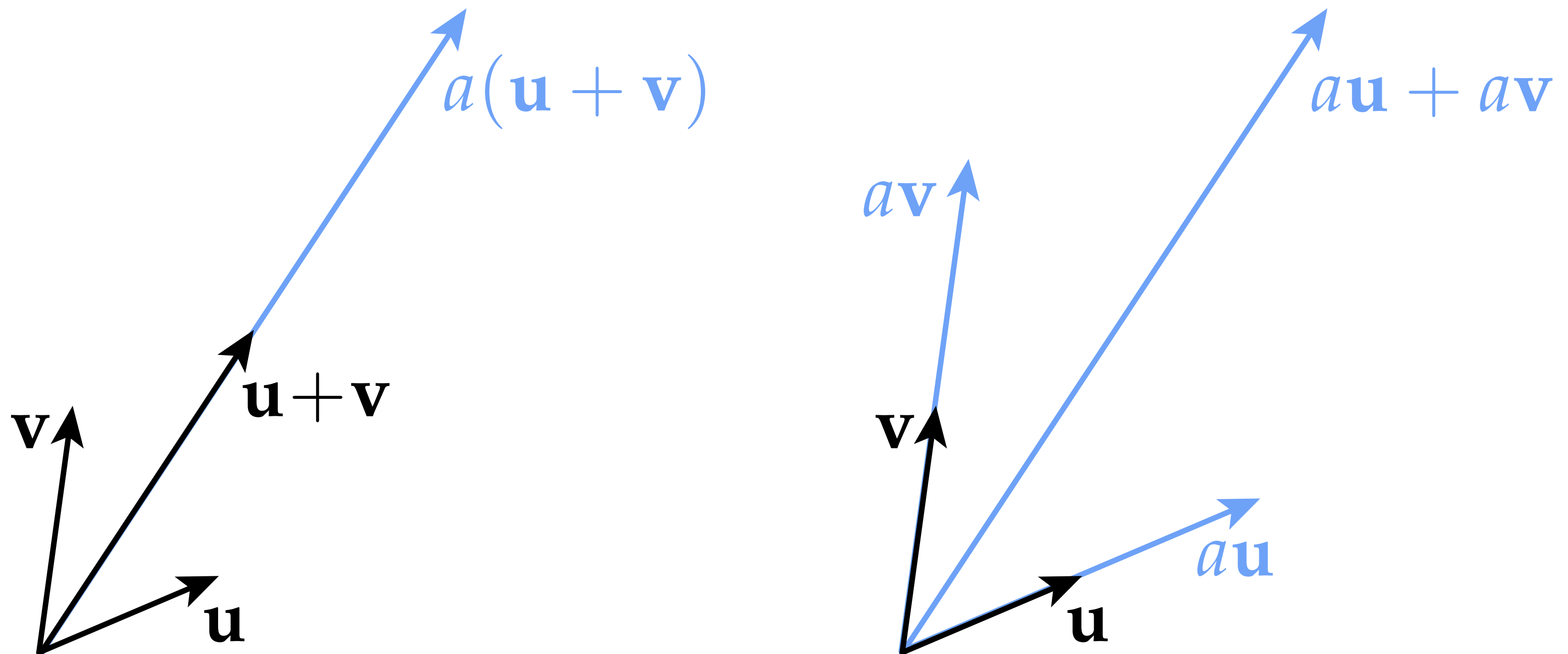


- In general, can multiply any vector u by a number or “scalar” a to get a new vector au .
- Multiplication behaves the way we would expect, based on the geometric behavior of scaling “little arrows.” E.g.,

$$a(bu) = (ab)u$$

Interaction of Addition & Scaling

- What if we try to add two scaled vectors? Or scale two vectors that have been added together?



- Interesting—seems we get the same result either way:

$$a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$$

Vector Space—Formal Definition

- If we keep playing around vectors, eventually we come up with a complete set of “rules” that vectors seem to obey:

For all vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ and scalars a, b :

- $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
- $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$
- There exists a *zero vector* “ $\mathbf{0}$ ” such that $\mathbf{v} + \mathbf{0} = \mathbf{0} + \mathbf{v} = \mathbf{v}$
- For every $\mathbf{v}$ there is a vector “ $-\mathbf{v}$ ” such that $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$
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- $a(b\mathbf{v}) = (ab)\mathbf{v}$
- $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$
- $(a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$

- ***These rules did not “fall out of the sky!”*** Each one comes from the geometric behavior of “little arrows.” (Can you draw a picture for each one?)
- ***Any* collection of objects satisfying all of this properties is a **vector space** (even if they don’t look like little arrows!)**

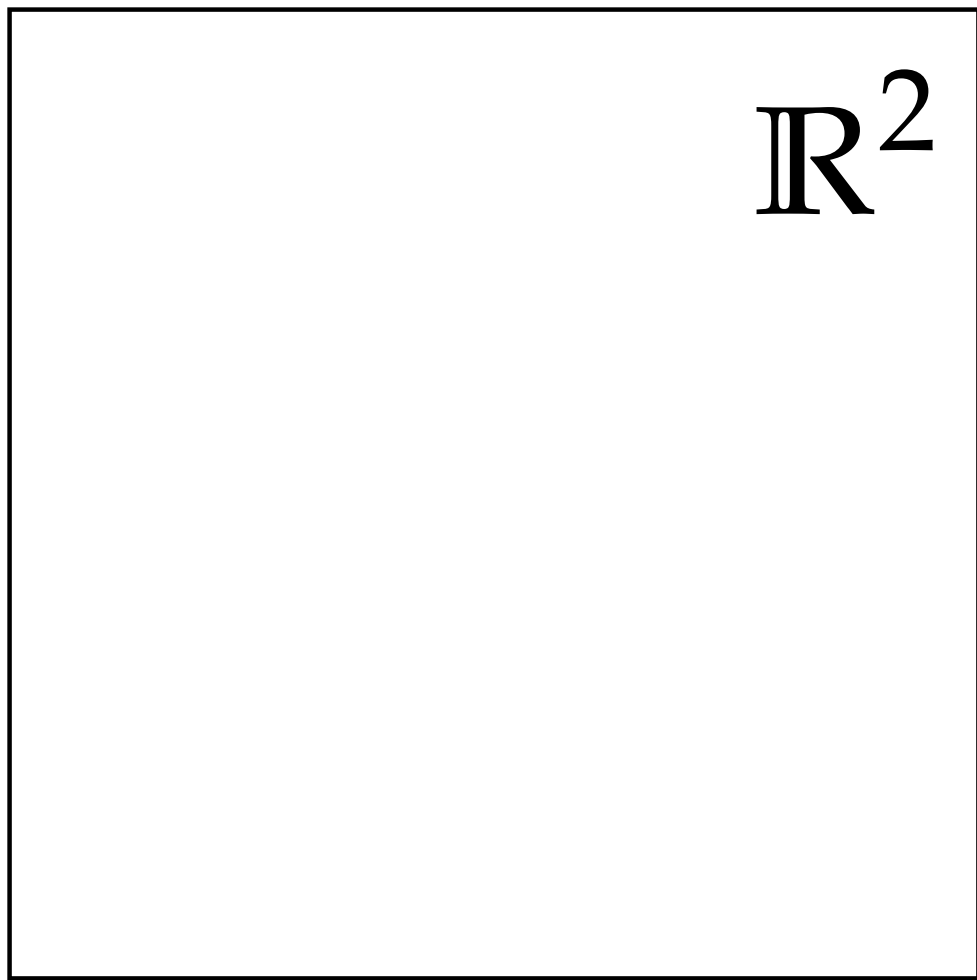
Euclidean Vector Space

- **Most common example: Euclidean n-dimensional space**
- **Typically denoted by $\mathbb{R}^n$, meaning “n real numbers”**
- **E.g., $(1.23, 4.56, \pi/2)$ is a point in $\mathbb{R}^3$**
- **Why such a common example?**
 - **Looks a lot like the space we live in!**
 - **That’s what we can easily encode on a computer (a list of floating-point numbers).**

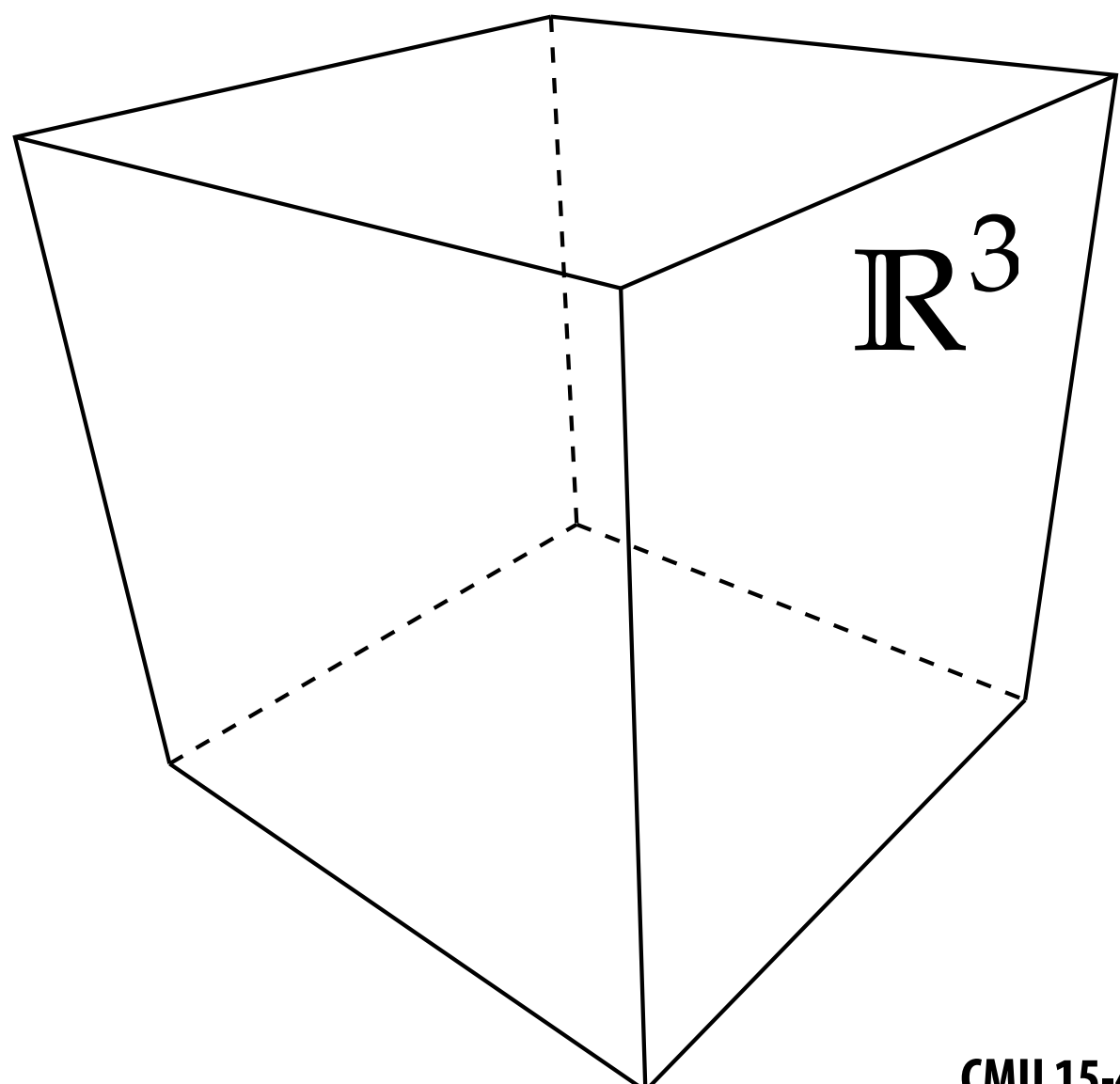
$\mathbb{R}$



$\mathbb{R}^2$

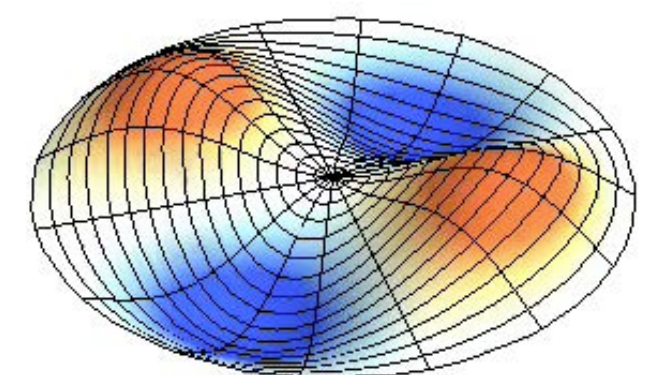
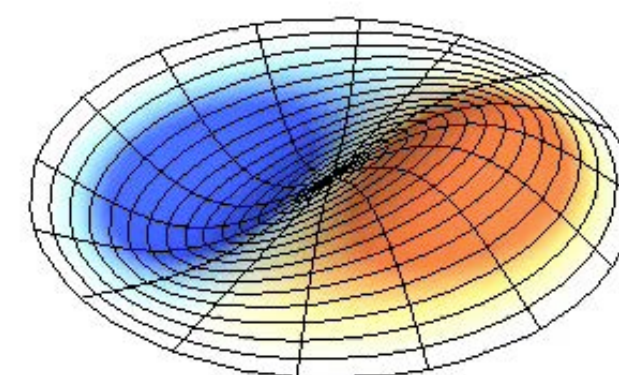
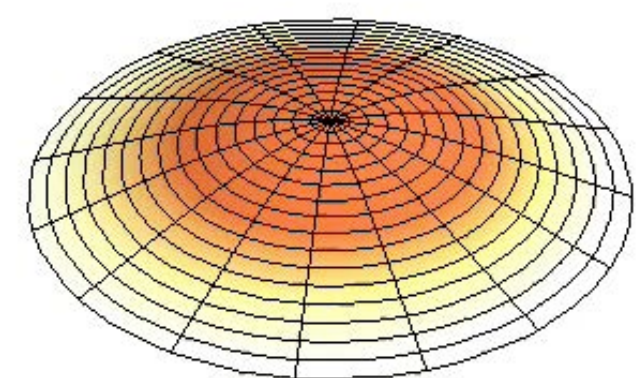
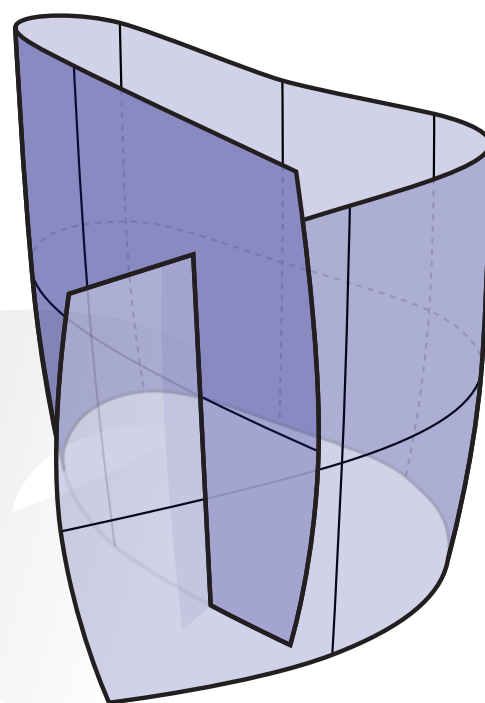
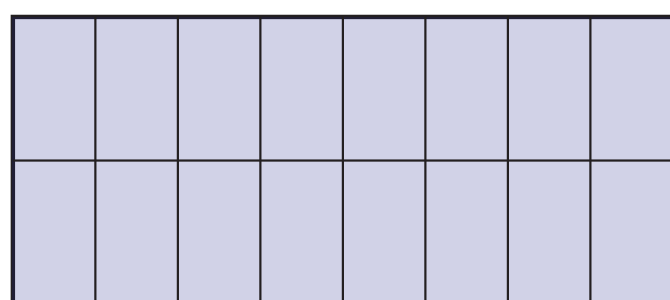
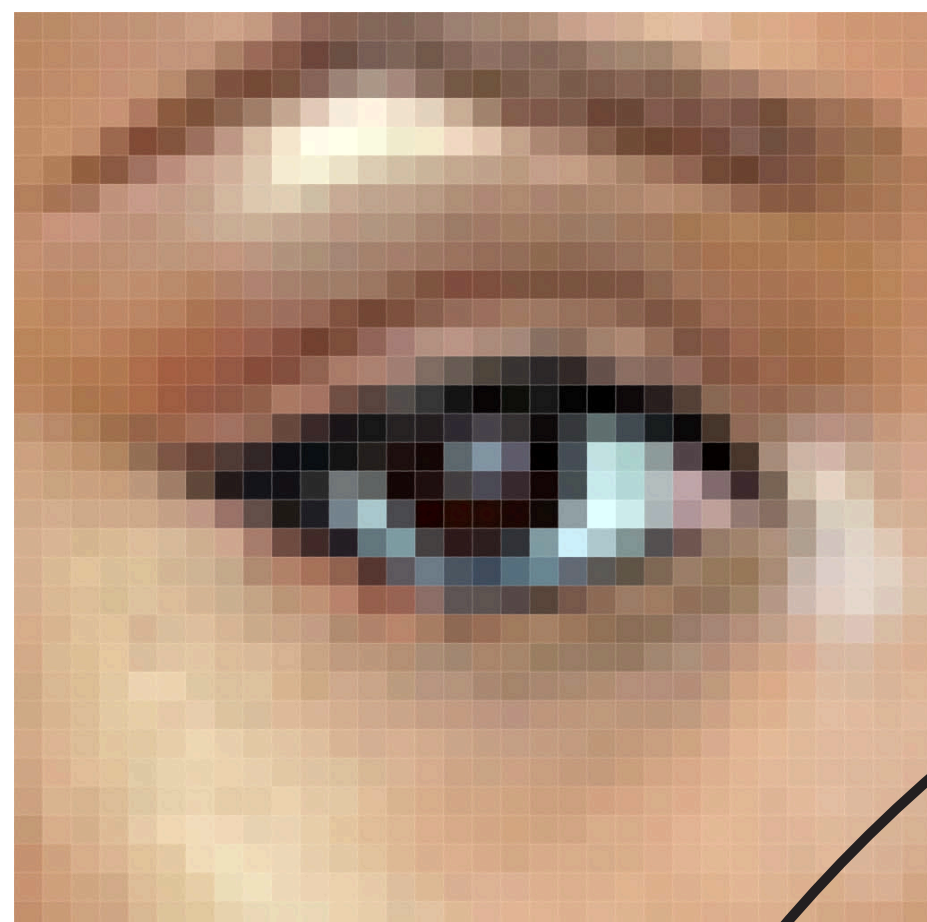


$\mathbb{R}^3$



Functions as Vectors

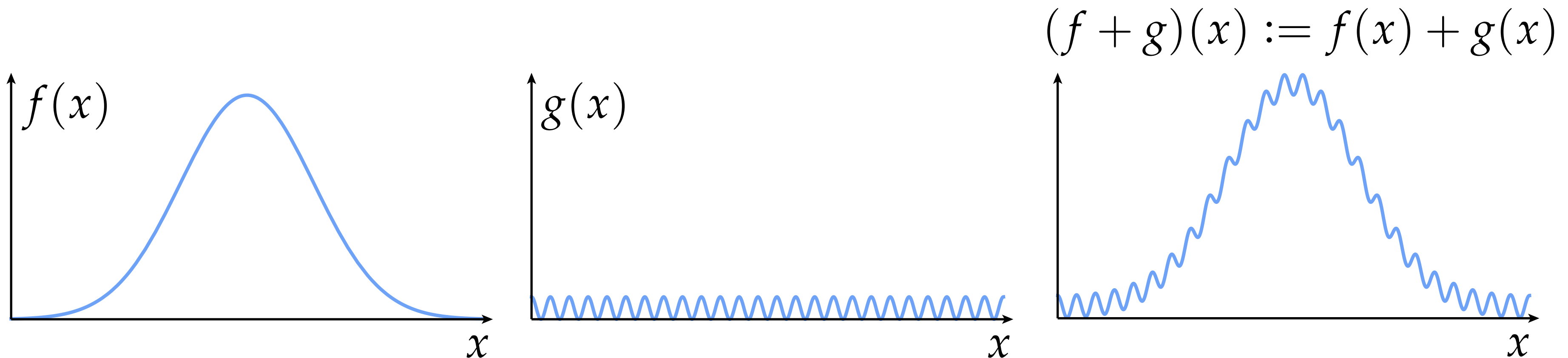
- Another very important example of vector spaces in computer graphics are spaces of *functions*.
- Why? Because many of the objects we want to work with in graphics are functions! (Images, radiance from a light source, surfaces, modal vibrations, ...)



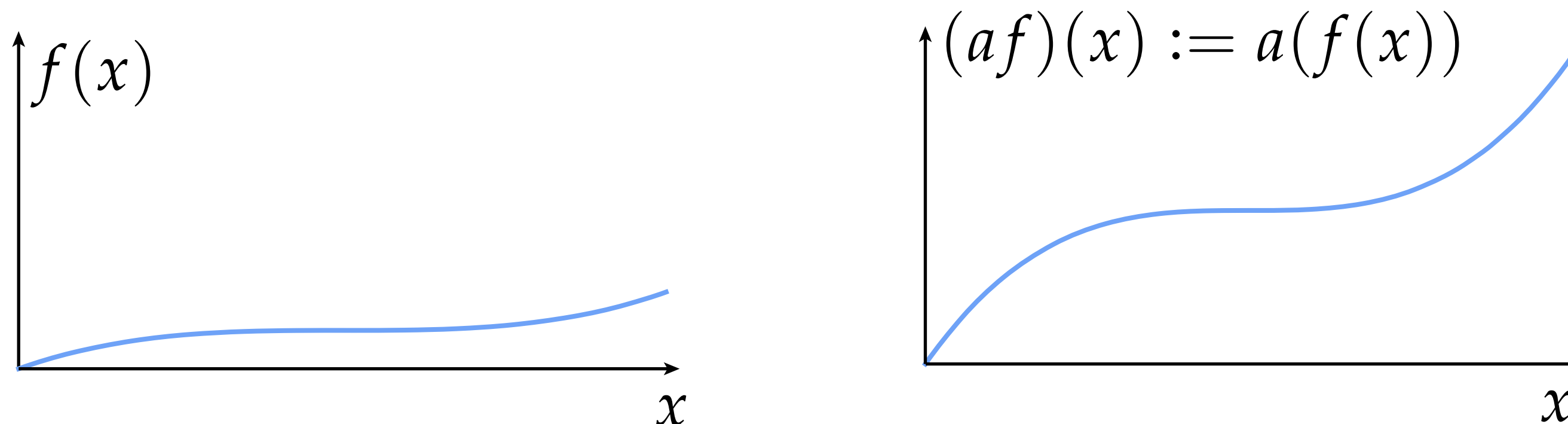
These are all vectors! :-)

Functions as Vectors

- Do functions exhibit the same behavior as “little arrows?”
- Well, we can certainly add two functions:



- We can also scale a function:



Functions as Vectors

■ What about the rest of these properties?

For all vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ and scalars a, b :

- $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
- $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$
- There exists a *zero vector* “ $\mathbf{0}$ ” such that $\mathbf{v} + \mathbf{0} = \mathbf{0} + \mathbf{v} = \mathbf{v}$
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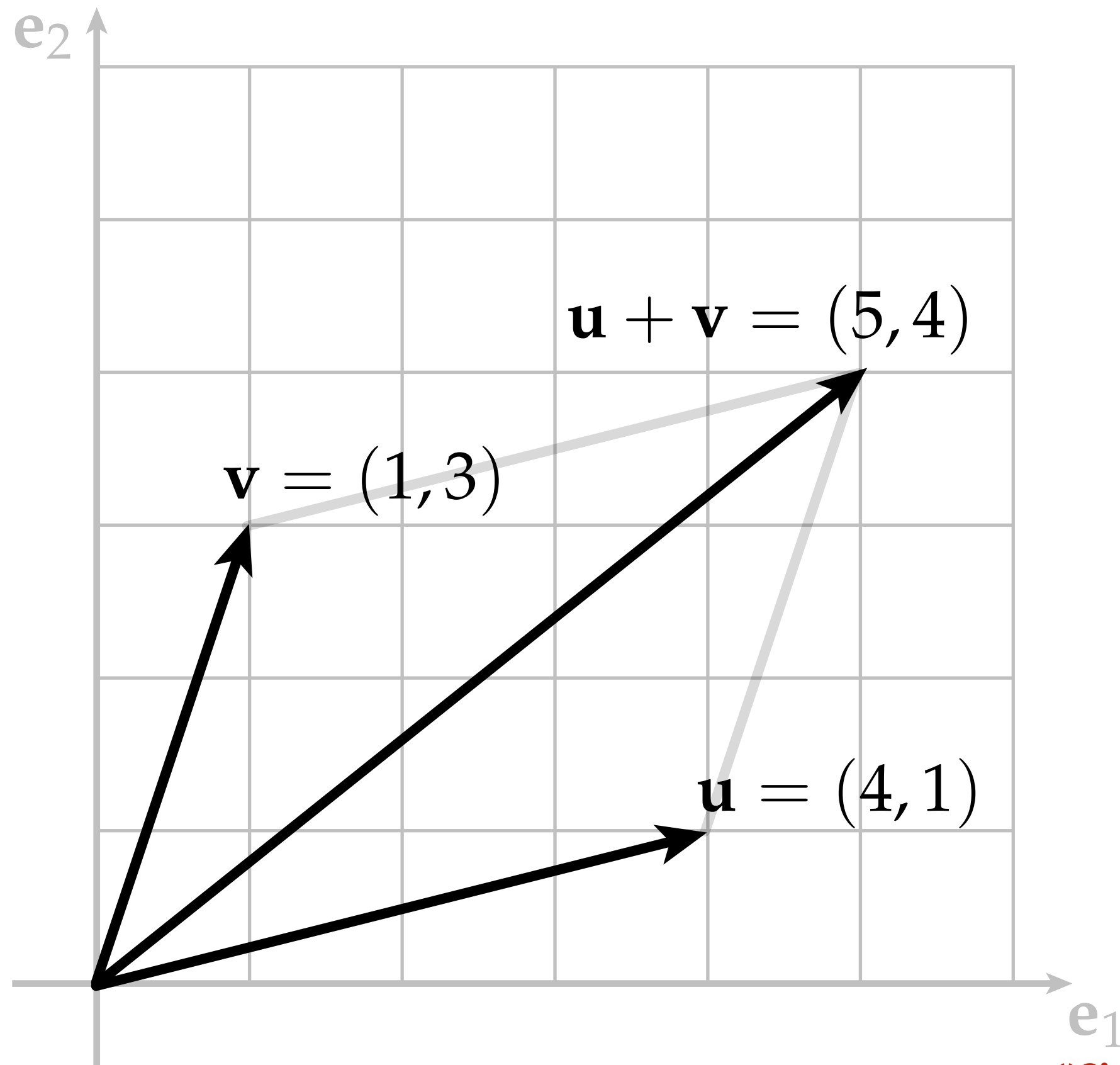
■ Try it out at home!

■ E.g., the “zero vector” is the function equal to zero for all x .

■ Short answer: yes, functions are vectors! (Even if they don’t look like “little arrows”.)

Vectors in Coordinates

- So far, we've only drawn our vector operations via pictures.
- How do we actually compute with vectors?
- Return to our coordinate representation:



$$\mathbf{u} + \mathbf{v}$$

$$= (1, 3) + (4, 1)$$

$$= (1 + 4, 3 + 1)$$

$$= (5, 4)$$

***Side note: does it make sense to add vectors encoded as (r, θ) ?**

**Ok, so we came up with some
rule for adding pairs of numbers.**

**How can we check that it faithfully encodes
geometric behavior of “little arrows?”**

From Geometry to Algebra

- Just check that it agrees with our list of rules that we know (from reasoning *geometrically*) “little arrows” must obey:

For all vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ and scalars a, b :

- $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
- $\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$
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- $a(b\mathbf{v}) = (ab)\mathbf{v}$
- $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$
- $(a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$

- For instance, for any two vectors $\mathbf{u} := (u_1, u_2)$ and $\mathbf{v} := (v_1, v_2)$ we have

$$\begin{aligned}\mathbf{u} + \mathbf{v} &= (u_1, u_2) + (v_1, v_2) = (u_1 + v_1, u_2 + v_2) = \\ &= (v_1 + u_1, v_2 + u_2) = (v_1, v_2) + (u_1, u_2) = \mathbf{v} + \mathbf{u}.\end{aligned}$$

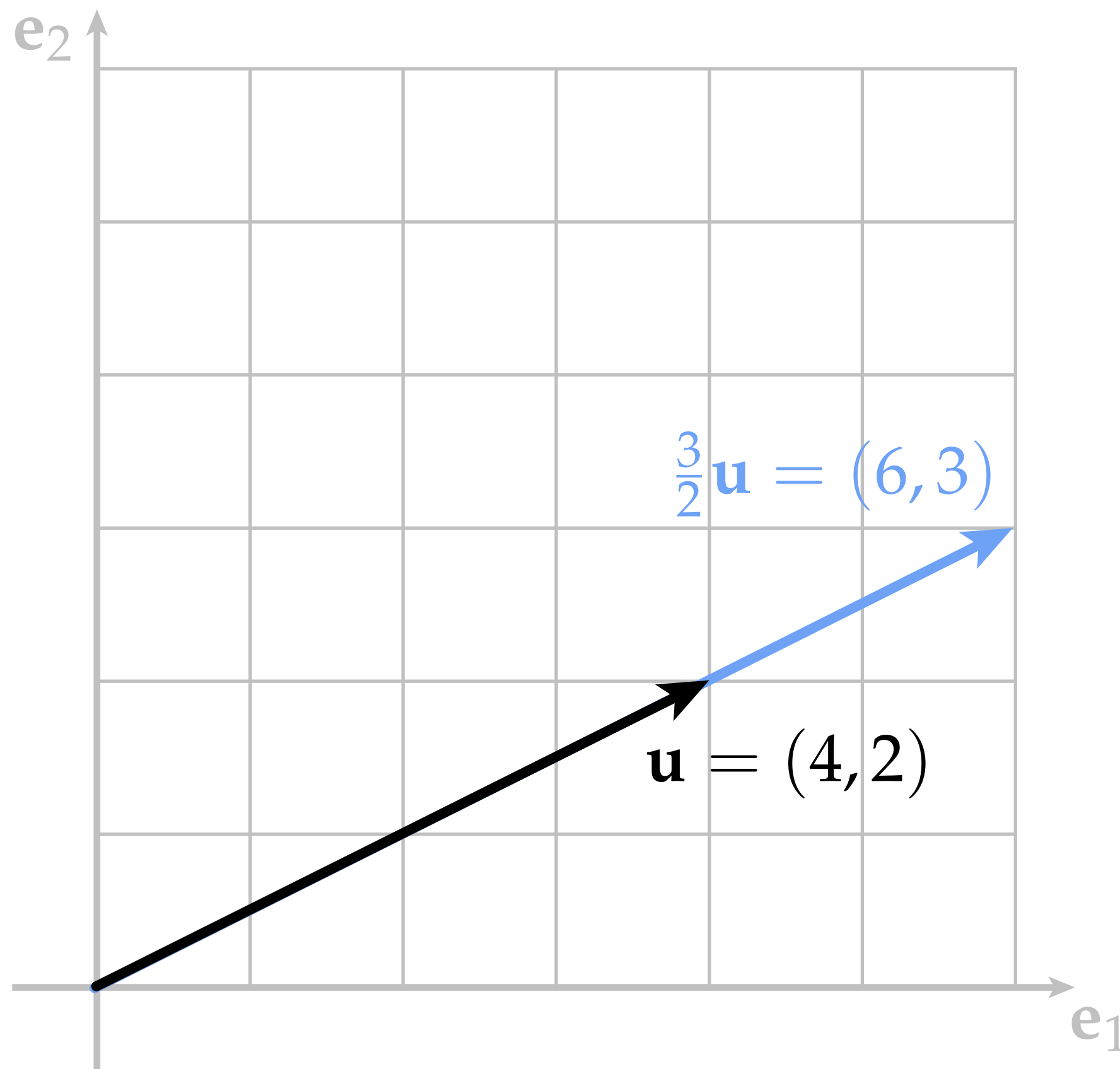
Turning geometric observations into algebraic rules is convenient for symbolic manipulation & numerical computation.

But you should *never* blindly accept a rule given by authority.

**Always ask: where does this rule come from?
What does it mean geometrically?
(Can you draw a picture?)**

Scaling Vectors in Coordinates

- We'd also like to be able to scale vectors using coordinates.
- Any ideas?



$$\frac{3}{2}\mathbf{u}$$

$$= \frac{3}{2}(4, 2)$$

$$= (4 \cdot 3/2, 2 \cdot 3/2)$$

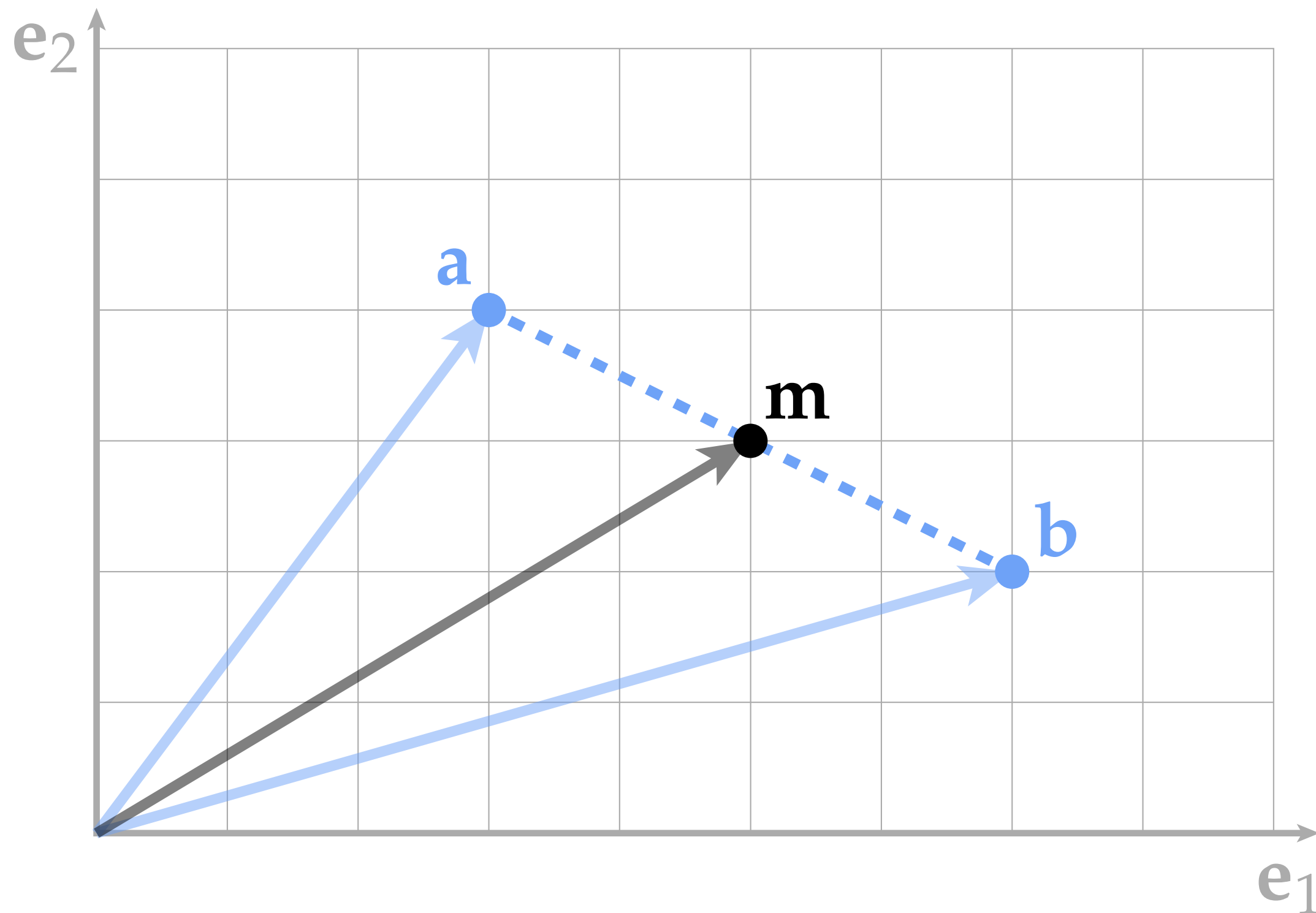
$$= (12/2, 6/2)$$

$$= (6, 3)$$

(From here, check the rest of the properties...)

Computing the Midpoint

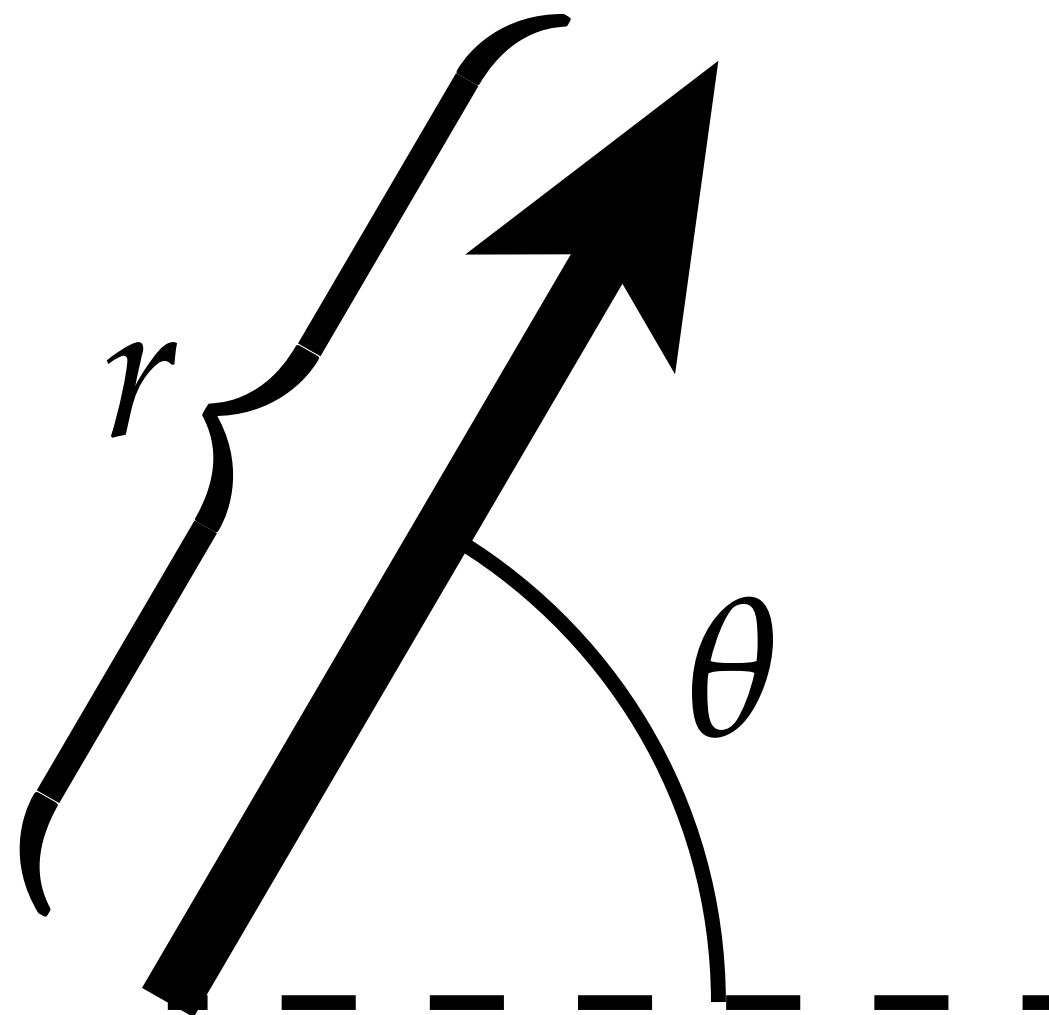
- As we start to combine vector operations, we build up operations needed for computer graphics.
- E.g., how would I compute the midpoint m of $a = (3,4)$ and $b = (7,2)$?



$$\begin{aligned} \mathbf{m} &= \frac{1}{2}(\mathbf{a} + \mathbf{b}) \\ &= \frac{1}{2}((3,4) + (7,2)) \\ &= \frac{1}{2}(10,6) \\ &= (5,3) \end{aligned}$$

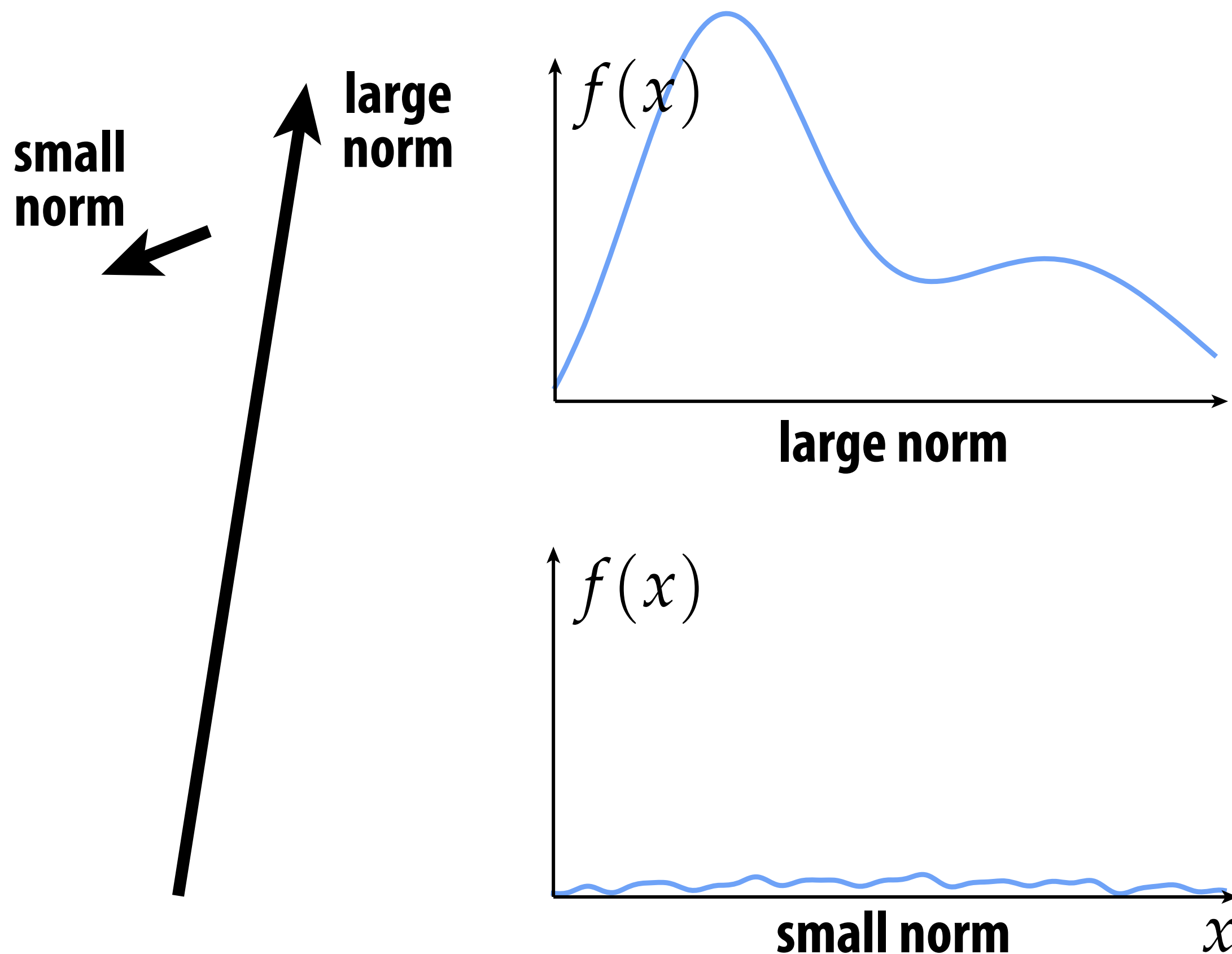
Measuring Vectors

- Earlier we asked, “what information does a vector encode?”
- (A: Orientation and magnitude.)
- How do we actually *measure* these quantities?

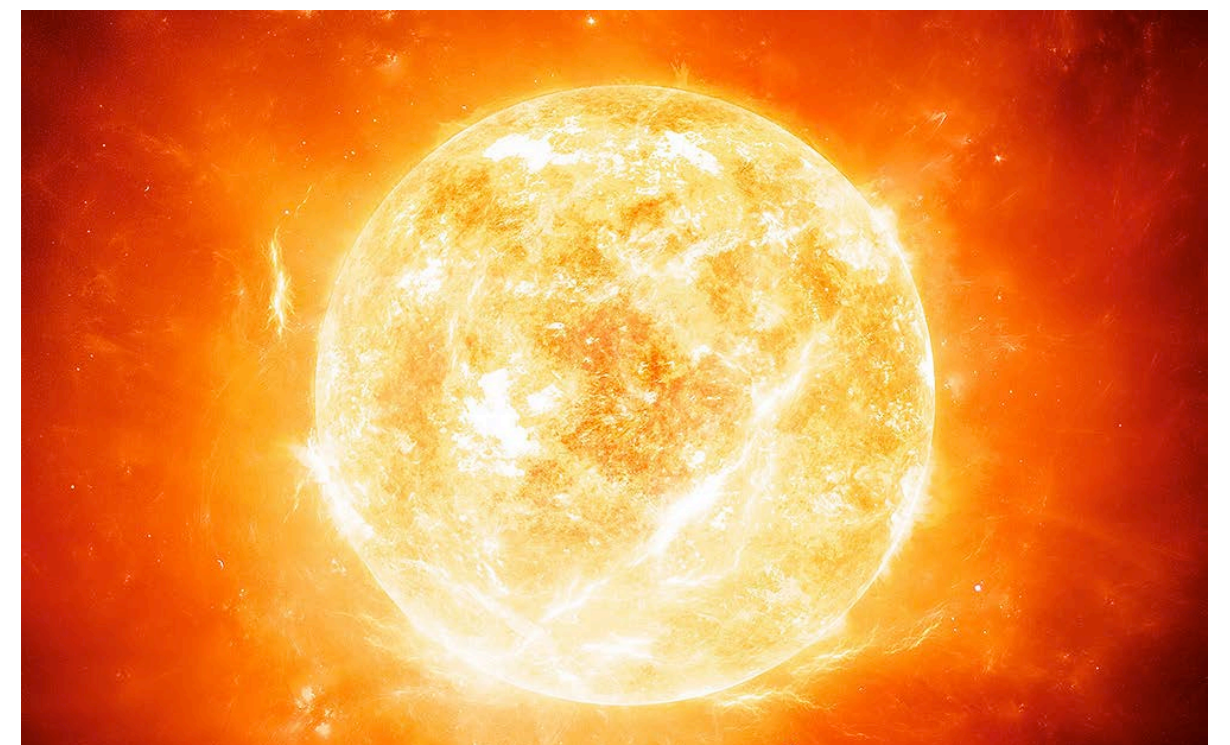


Norm of a Vector

- Let's start with magnitude—for a given vector v , we want to assign it a number $|v|$ called its **length** or **magnitude** or **norm**.
- Intuitively, the norm should capture how “big” the vector is.



small norm



large norm

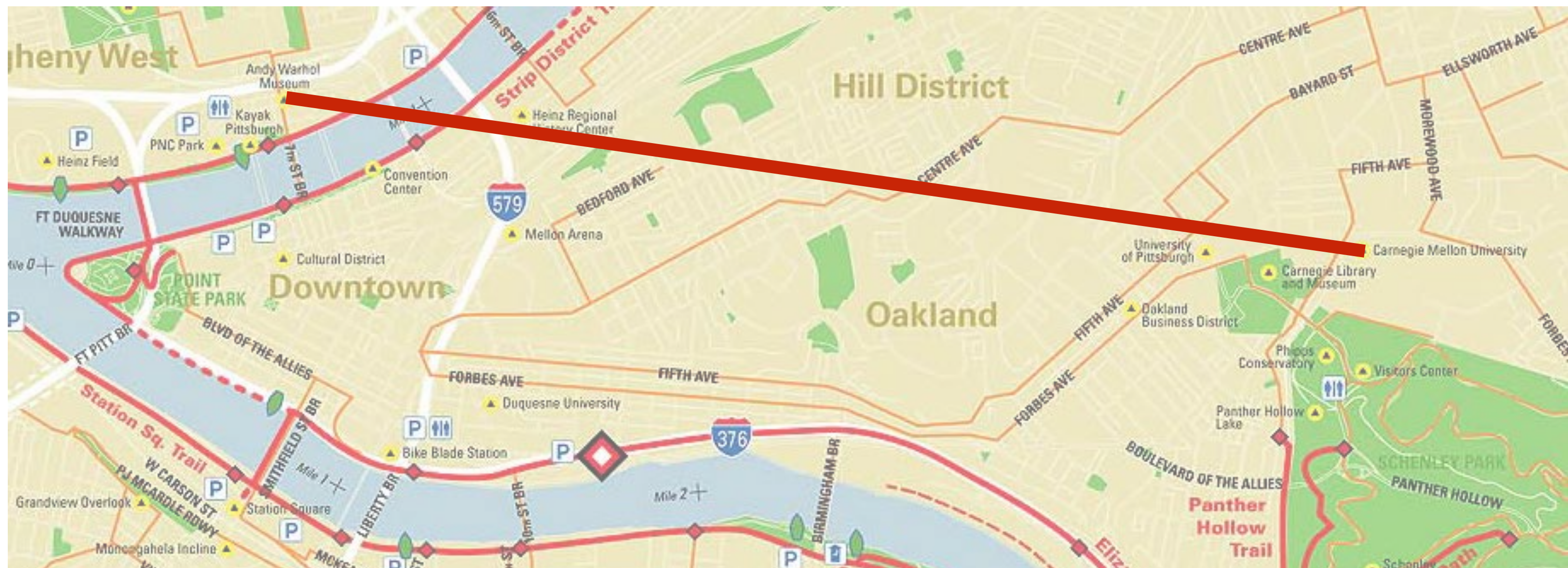
Natural Properties of Length—Positivity

- What properties might you expect the norm (or length) of a vector to satisfy?
- For one thing, it probably shouldn't be negative!

$$|\mathbf{u}| \geq 0$$

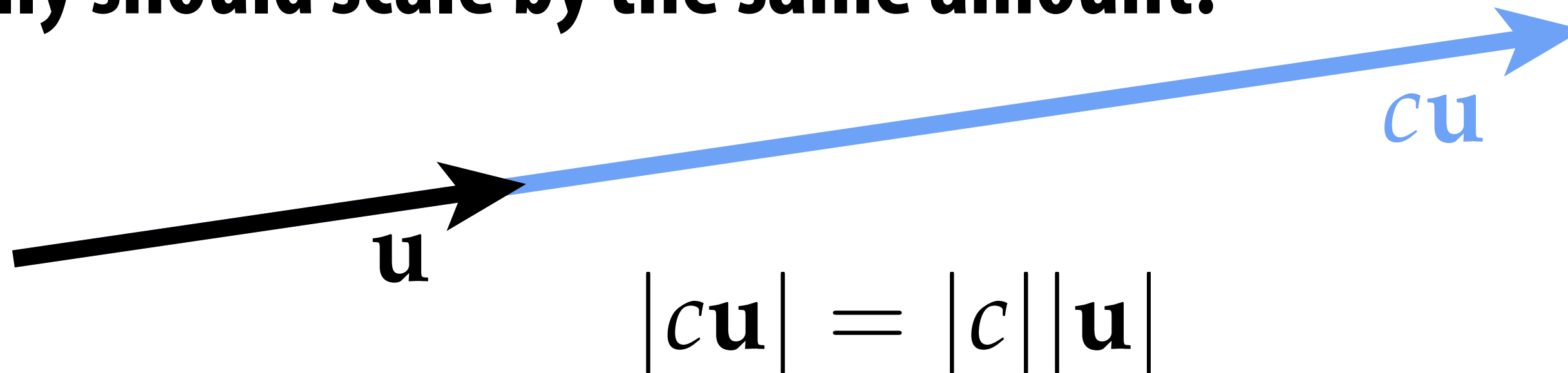
- And probably it should be zero only for the zero vector:

$$|\mathbf{u}| = 0 \iff \mathbf{u} = \mathbf{0}$$

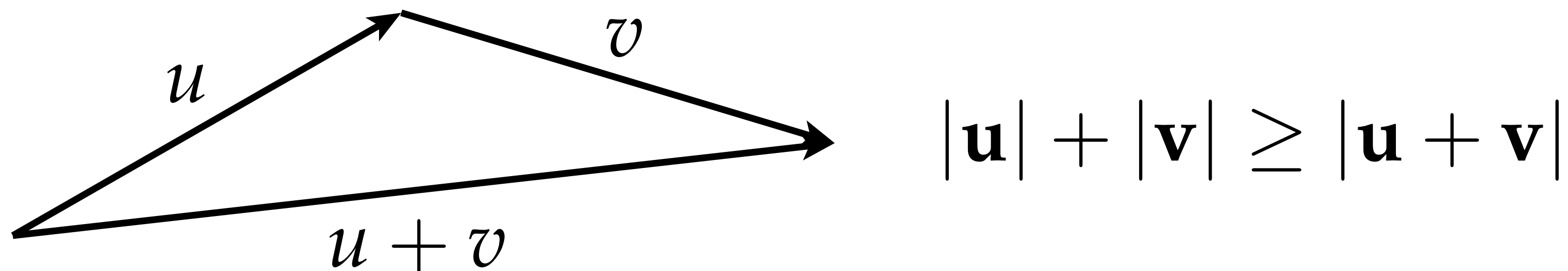


Natural Properties of Length, Continued

- Also, if we scale a vector by a factor c , its norm (i.e., length) really should scale by the same amount:



- Finally, we know that the shortest path between two points is always along a straight line:



- (This final property is sometimes called the “pentagon inequality,” since the diagram looks like a pentagon.)

Norm—Formal Definition

- A norm is any function that assigns a number to each vector and satisfies the following properties for all vectors $\mathbf{u}$, $\mathbf{v}$, and all scalars a :

- $|\mathbf{v}| \geq 0$

- $|\mathbf{v}| = 0 \iff \mathbf{v} = \mathbf{0}$

- $|a\mathbf{v}| = |a||\mathbf{v}|$

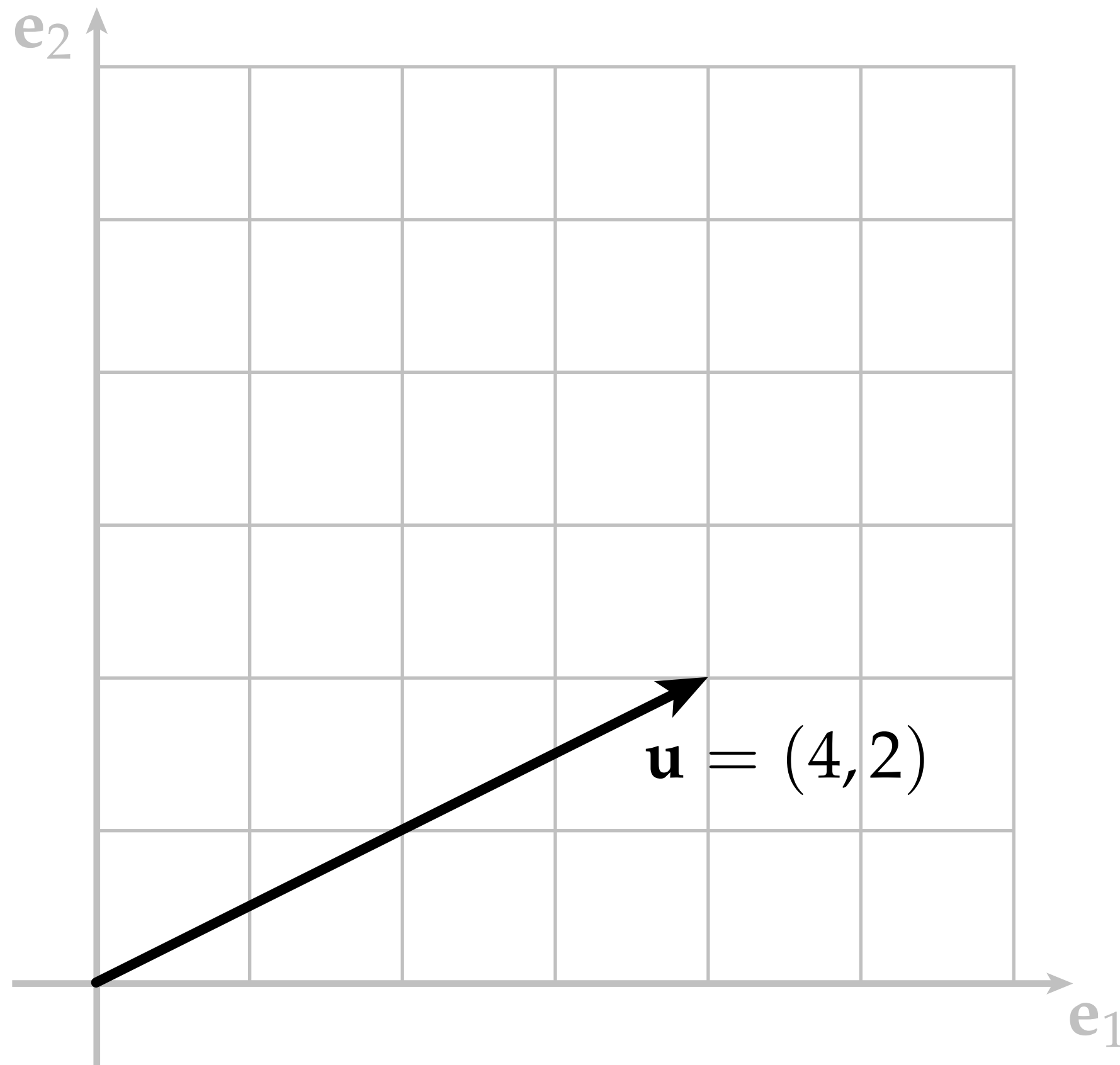
- $|\mathbf{u}| + |\mathbf{v}| \geq |\mathbf{u} + \mathbf{v}|$

- But you don't have to take my word for it—for each rule, you now have a concrete geometric picture explaining why this “rule” is there.

Euclidean Norm in Cartesian Coordinates

- A standard norm is the so-called *Euclidean norm* of n -vectors:

$$|\mathbf{u}| = |(u_1, \dots, u_n)| := \sqrt{\sum_{i=1}^n u_i^2}$$



Example: $\mathbf{u} = (4, 2)$

$$\begin{aligned} |\mathbf{u}| &= \sqrt{4^2 + 2^2} \\ &= 2\sqrt{5} \end{aligned}$$

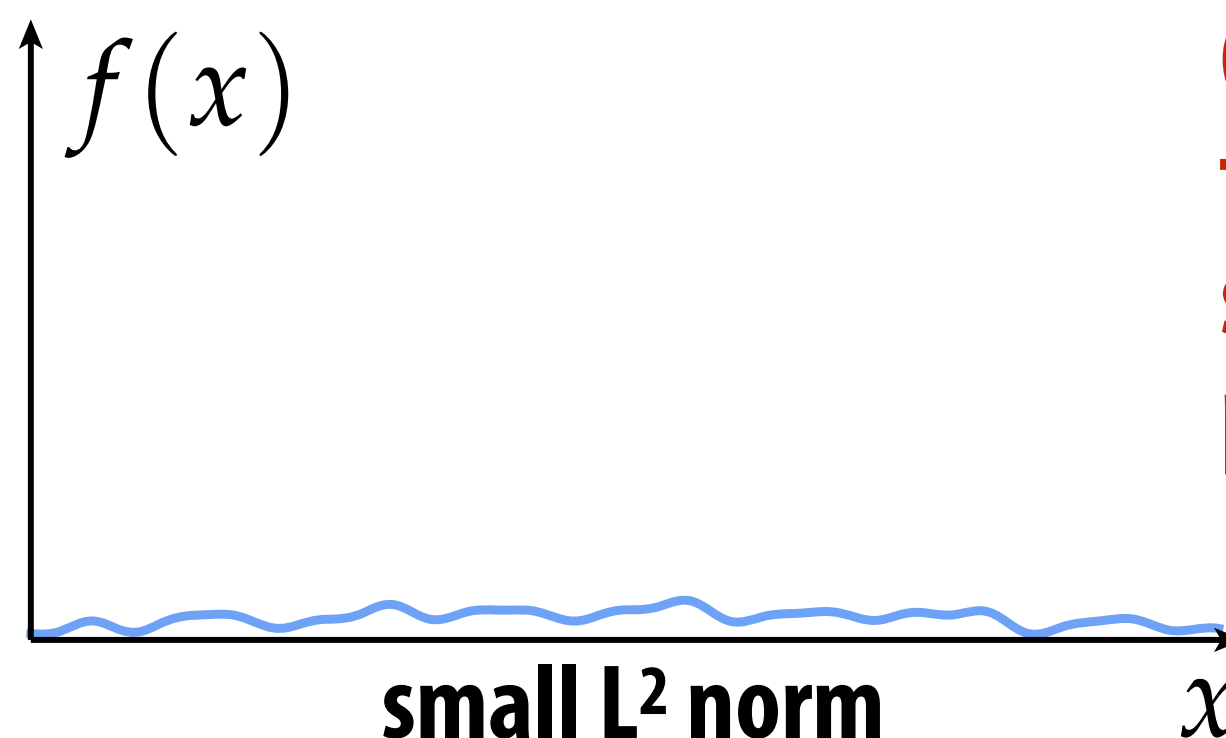
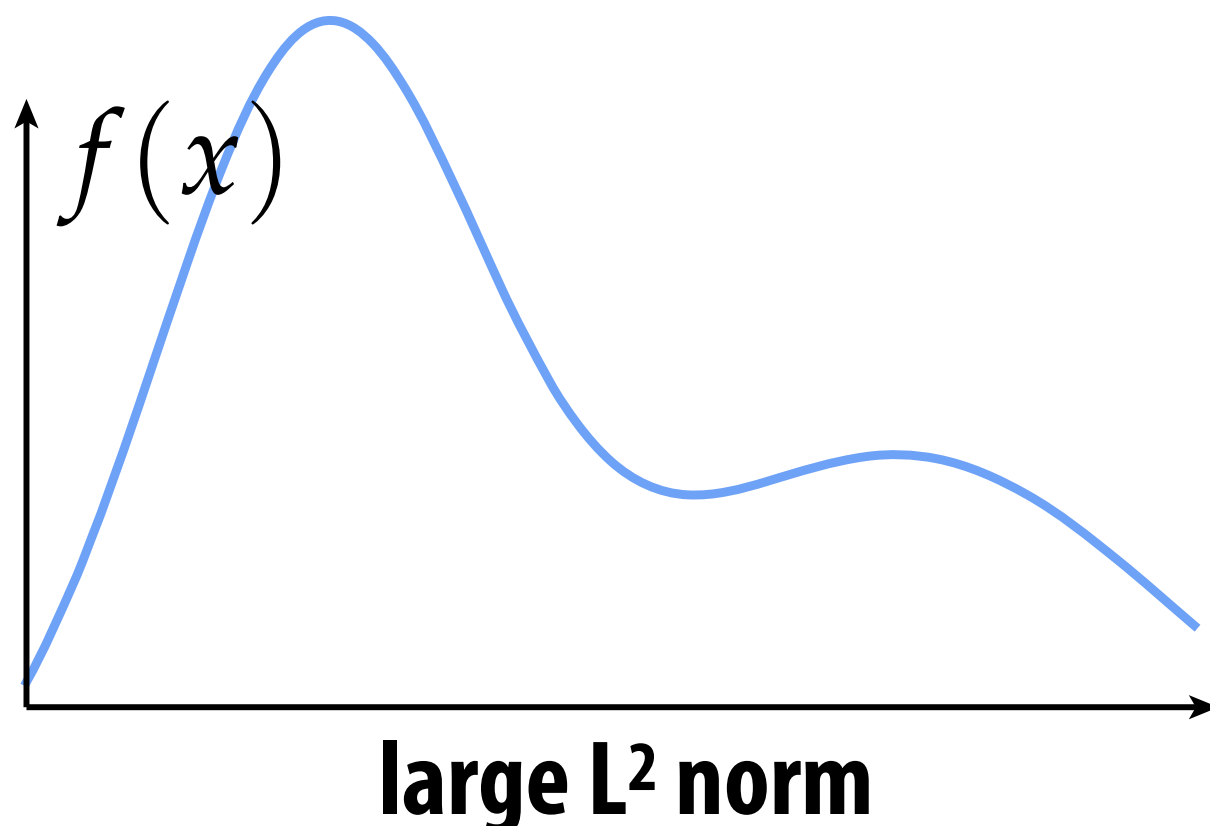
**Q: Does this formula satisfy all the natural, geometric properties of a norm?
(Answer in the slide comments!)**

L² Norm of Functions

- Less familiar idea, but same basic intuition: the so-called *L² norm* measures the total magnitude of a function.
- Consider real-valued functions on the unit interval [0,1] whose square has a well-defined integral. The L² norm is defined as:

$$||f|| := \sqrt{\int_0^1 f(x)^2 dx}$$

- Not too different from the Euclidean norm: we just replaced a sum with an integral (which is kind of like a sum...).



Q: Careful—does the formula above *exactly* satisfy all our desired properties for a norm?

L² Norm of Functions—Example

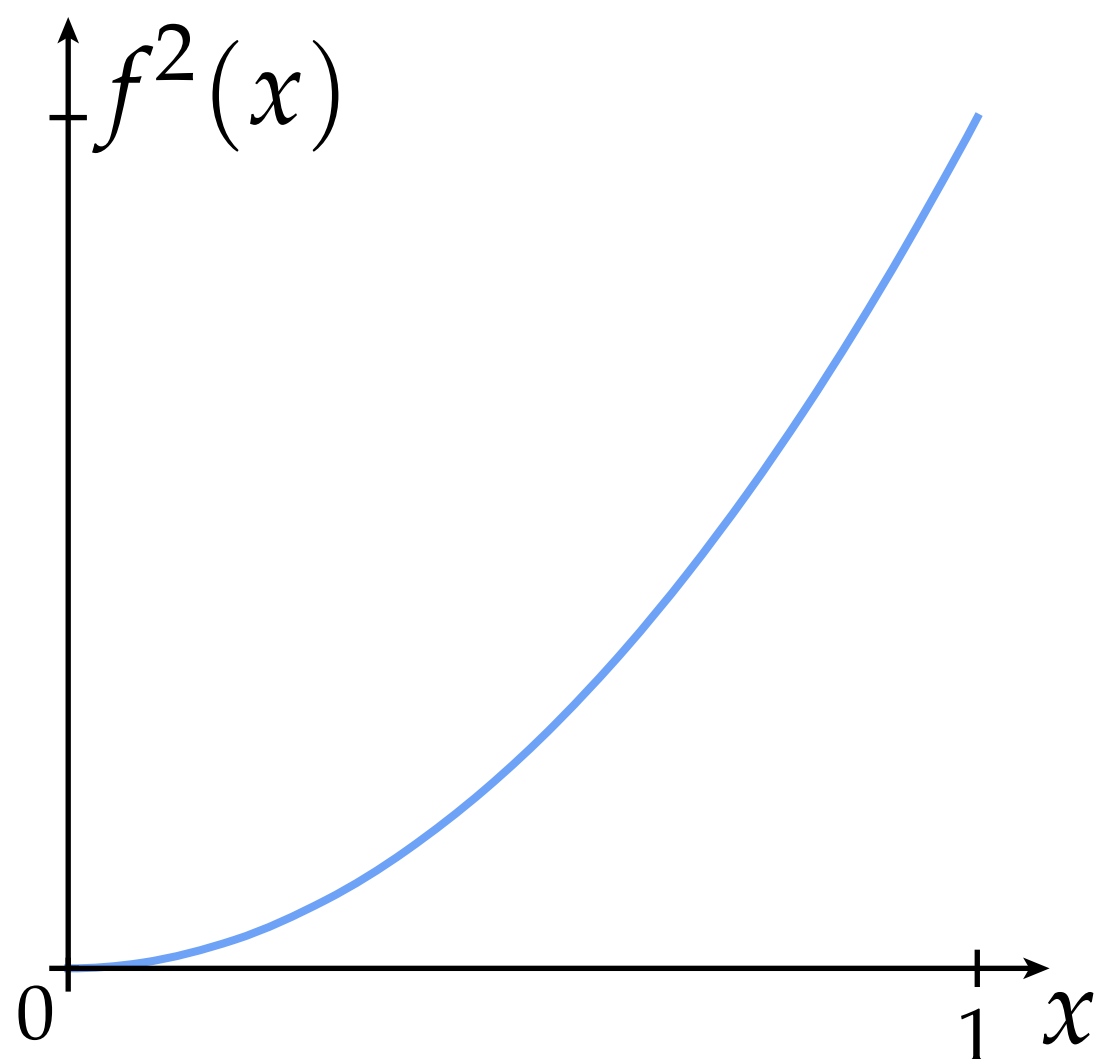
- Consider the function $f(x) := x\sqrt{3}$, defined over the unit interval $[0,1]$. $\|f\| := \sqrt{\int_0^1 f(x)^2 dx}$

- Q: What is its L² norm?

- A:

$$\|f\|^2 = \int_0^1 3x^2 dx = \left[x^3 \right]_0^1 = 1^3 - 0^3 = 1$$

$$\Rightarrow \|f\| = \sqrt{1} = 1.$$



For clarity we will use $\|\cdot\|$ for the norm of a function, and $|\cdot|$ for the norm of a vector in $\mathbb{R}^n$.

P.S. Most integrals in graphics are not calculated this way (at least not for more challenging functions, or functions described by data). Later on we'll talk a *lot* about numerical integration.